- *System 2*: Processes are calculated and analytical.

System 1 thinking, while imperfect, is speedy and highly efficient. For example, if Joe is facing the threat of a large tiger charging him at full speed, Joe's System 1 thinking will trigger him to turn around and sprint for the nearest tree, and ask questions later. As an alternative, Joe's System 2 thinking will calculate the speed of the tiger's approach and assess his situation. Joe will examine his options and realize that there is an armored vehicle a few steps away. On average, if Joe immediately sprints to the tree he may get lucky and outrun the tiger; on the other hand, if Joe stops, thinks through his tiger-escaping options, and rationally selects the optimal choice for escape, well, there may be one less Joe in the jungle…

Joe's hypothetical situation highlights why evolution has created System 1; on average, running for the tree is a simple and reliable life-saving decision, versus the more complex and time-consuming alternative of thinking through the options and making a so-called "rational" decision. While System 1 is uber-efficient and fast, the downside of System 1 is that its "fight or flight" nature often leads to systematic bias: Joe will *always* run for the tree, even when the better decision may be to stop, think about alternatives, and jump in the armored vehicle.

System 1 certainly served its purpose when humans were faced with life and death situations in the jungles. In modern-day life, where saber-tooth tigers are less common and long-term consequences are the norm, the benefits of immediate decisions rarely outweigh the costs of flawed decision-making. In finance, the necessity of avoiding System 1 and relying on System 2 in the context of navigating financial markets is of utmost importance.

Below are three core reasons why human experts underperform systematic simple models, with each cause discussed in more detail in the following chapters.

1. Same Facts; Different Decisions

Humans, unlike models, can take the same set of facts and arrive at different conclusions. This can happen for a variety of reasons, but a lack of human consistency is often attributed to reacting to what you first hear